

OpenClaw AI Rock-Paper-Scissors Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on the AI rock-paper-scissors project itself.

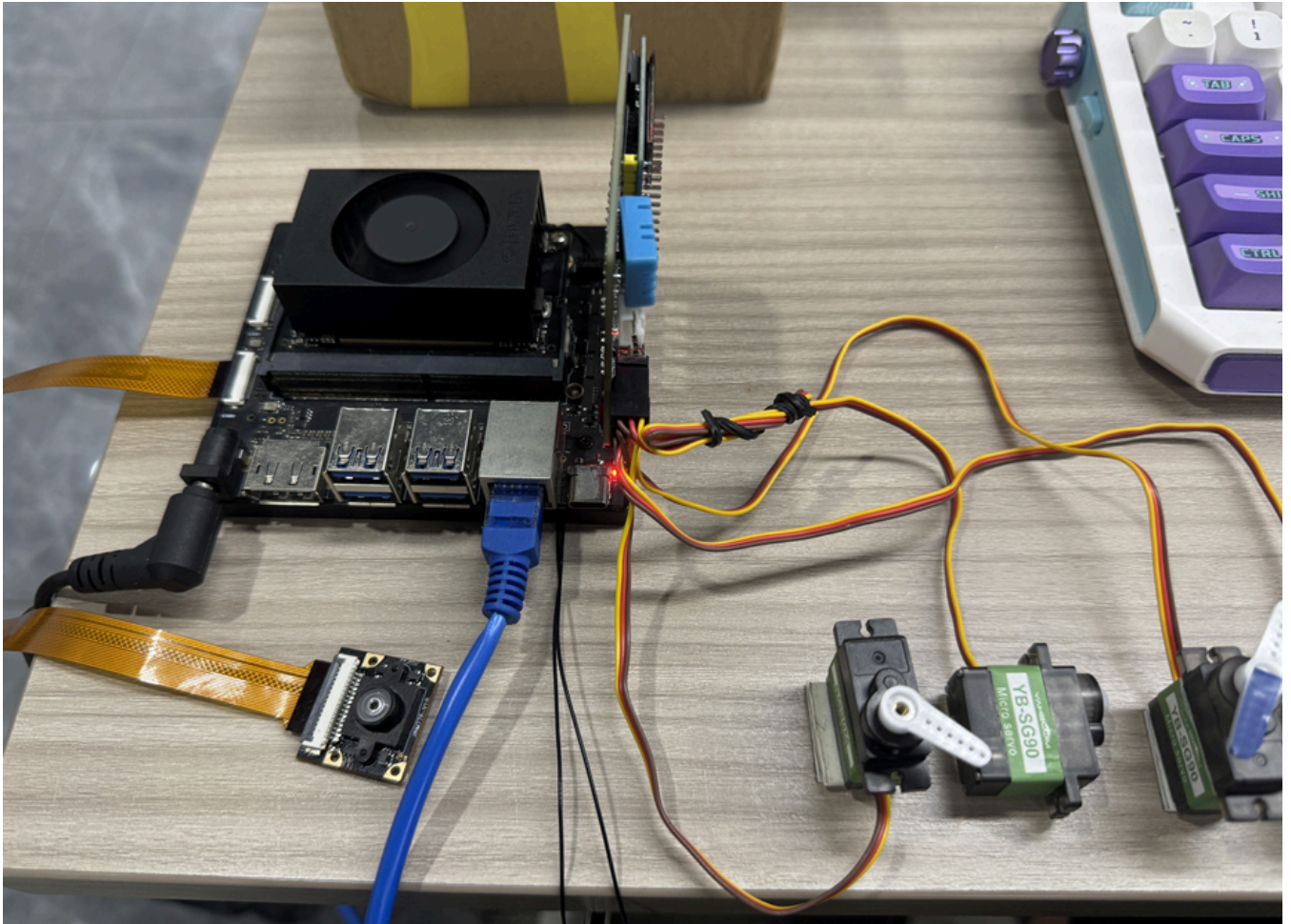
1. Tutorial Objectives

This article demonstrates OpenClaw's interactive gaming capability through "gesture recognition + AI referee + screen and light feedback". After completing this tutorial, readers can:

- Understand how to use the camera to recognize player gestures
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four project scenarios: gesture recognition, AI move, result judgment, and score feedback

2. Hardware Preparation

- OpenClaw mainboard
- CSI camera module
- OLED display
- RGB light strip / RGB light bead module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Camera capture command: `/home/jetson/hardware_hal_demo/seewhat`
- Voice entry: `/home/jetson/voice_to_openc1aw/src/main.py`

The key capabilities of this project mainly consist of three parts:

- The camera is responsible for capturing player gestures
- OpenClaw is responsible for deciding AI's move and displaying the result on OLED / RGB

4. Test Basic Project Capabilities

Before starting the current project, please complete a `openc1aw tui` basic dialogue self-check first; if you haven't done so, please refer to [05-Three Dialogue Solution Configuration Tests]. After passing, use TUI to verify the basic capabilities in the project.

Enter in terminal:

```
openc1aw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then you can do a few basic tests:

- Take a gesture photo for me
- Identify whether this is a peace sign
- Display "Ready" on OLED
- Light up the RGB in blue

5. Three Dialogue Solutions with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Use
whatsapp	Remote demo, event interaction	Easy to share with multiple users	Demonstrate "visual interactive games"
TUI	Local debugging, rule verification	Easiest to see recognition and judgment process	Run through game logic first
Voice Code	Game interaction, classroom demo	More like a real AI gaming partner	As a complete demonstration solution

If you have opted for the AI voice module + speaker, and whatsapp dialogue, TUI, or WhatsApp dialogue needs to broadcast OpenClaw's reply content, you can run the following program in the terminal. AI voice module dialogue does not need to run this repeatedly

- Modify the parameter for enabling/disabling broadcast

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast. After modification, save and run the following command in terminal
```

- Enter in terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Solution

For whatsapp integration and bot creation steps, please refer directly to [05-Three Dialogue Solution Configuration Tests]. After setup, you can remotely trigger the rock-paper-scissors game in WhatsApp, for example:

- Please play rock paper scissors. Take a photo to recognize my gesture. You can randomly make a gesture. Then judge whether you win or I win OLED displays who this round

20:27



< 2

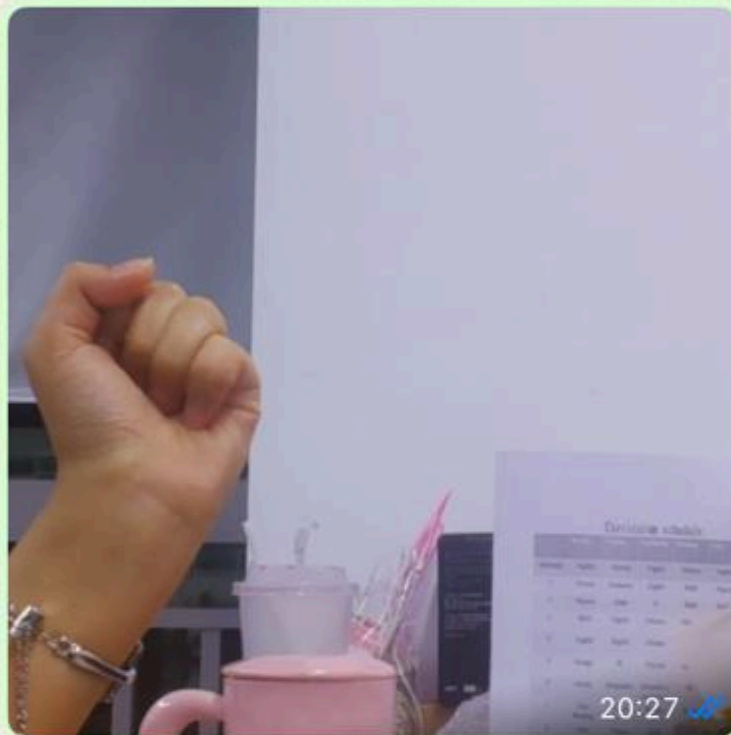


+86 153 3885 7526 (你)

给自己发消息

Please play rock paper scissors.
Take a photo to recognize my
gesture. You can randomly make a
gesture. Then judge whether you win
or I win OLED displays who this round.

20:27 ✓✓



I can see the user's hand gesture in
the image - it's a **fist**, which
represents **ROCK** in rock paper
scissors.

Now I need to:

1. Make a random gesture choice
(rock, paper, or scissors)
2. Determine the winner
3. Display the result on OLED





You can add your own whatsapp conversation screenshots here.

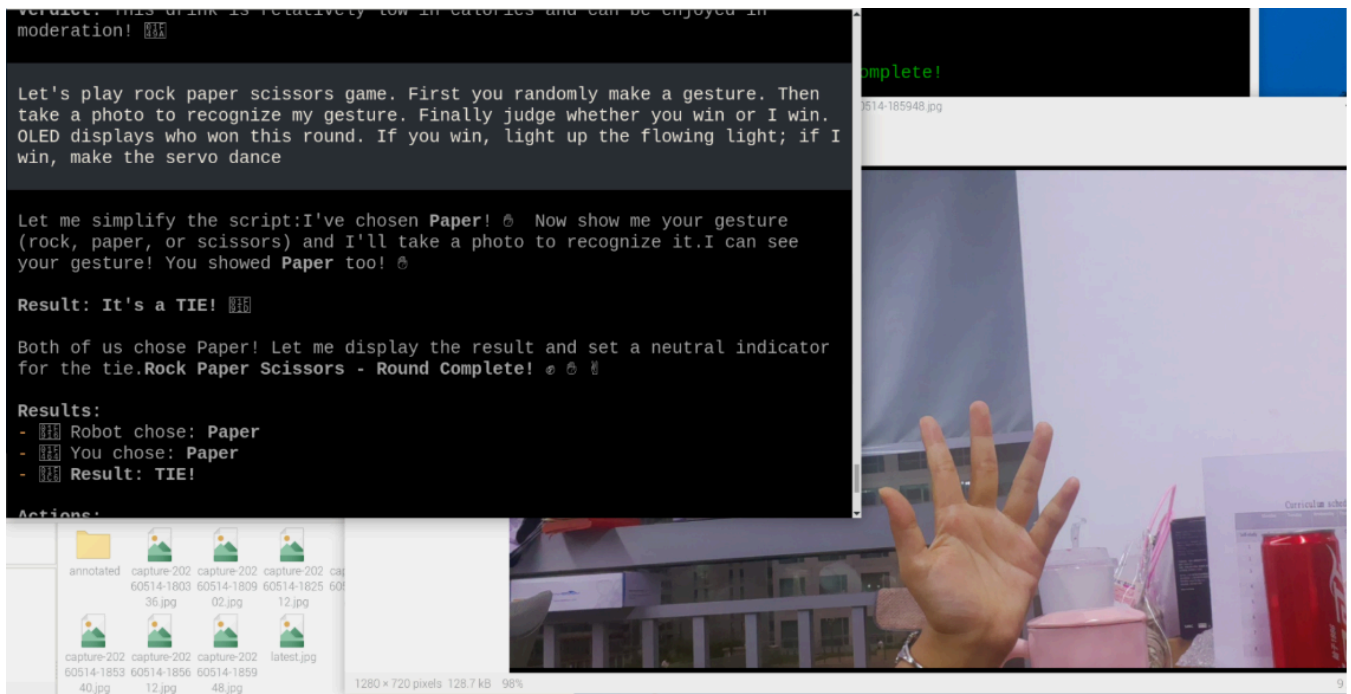
5.2 TUI Solution

If you have completed the TUI self-check in the prerequisites, directly enter terminal text dialogue:

```
openc1aw tui
```

Then enter project-related commands, for example:

- Let's play rock paper scissors game. First you randomly make a gesture. Then take a photo to recognize my gesture. Finally judge whether you win or I win. OLED displays who won this round. If you win, light up the flowing light; if I win, make the servo dance

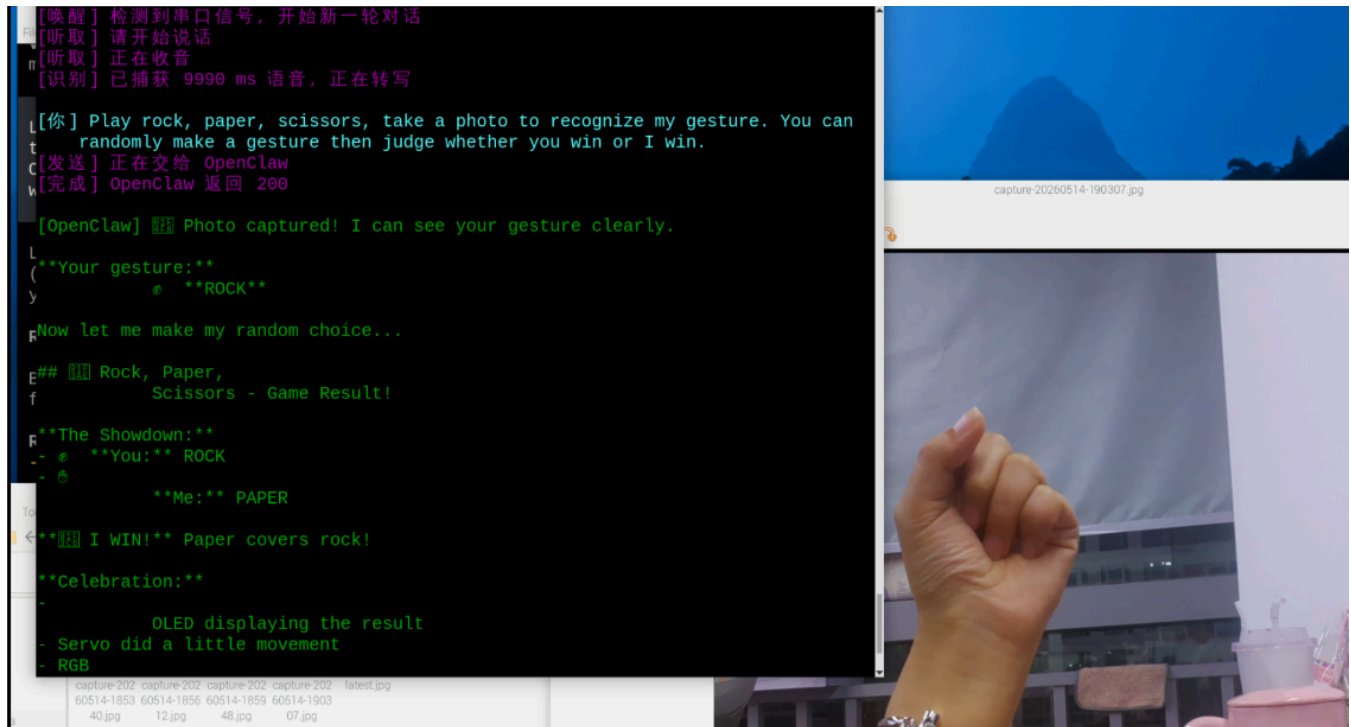


5.3 Voice Code Solution

Requires the optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [04-AI Voice Interaction Configuration]. After wake-up, you can directly speak the control content, for example:

- Let's play rock, paper, scissors, take a photo to recognize my gesture. You can randomly make a gesture then judge whether you win or I win.



6. Project Implementation

Note: The following demonstrations are all done through `openc1aw tui` dialogue. You can also choose whatsapp or voice methods to complete them.

6.1 Scenario 1: Gesture Recognition

Experiment Objective

Recognize the player's current gesture, such as rock, paper, or scissors.

Effect Description

OpenClaw will first take a photo, then identify the gesture type based on the image. This step is the input foundation for the entire game.

Example Commands

- Identify the gesture I'm making right now
- I've already made my move, help me check
- See if this is rock, paper, or scissors

6.2 Scenario 2: AI Move and Judgment

Experiment Objective

After recognizing the player's gesture, have OpenClaw provide the AI's move and judge the winner.

Effect Description

This step best demonstrates the "visual input + conversational reasoning" feature. OpenClaw not only recognizes the player but also continues to participate in the game, providing its own move and result explanation.

Example Commands

- Let's play rock-paper-scissors, take a photo to recognize my gesture, you can randomly make a gesture, then judge whether you won or I won, OLED displays who won this round

6.3 Scenario 3: OLED Scoreboard

Experiment Objective

Have OLED display the current round result or cumulative score.

Effect Description

The screen can display `win`, `Lose`, `Draw`, or show a simple score like `You 1 : 0 AI`, making the project feel more like a complete mini-game.

Example Commands

- Display this round's result on OLED
- Write the score to the screen
- Display "Round 1 You Win"

6.4 Scenario 4: RGB Result Feedback

Experiment Objective

Use RGB colors to quickly express win/loss results.

Effect Description

For example, green light for win, red light for loss, blue light for draw. This way, even without reading text, you can immediately know the result of the round.

Example Commands

- If I win, light up green
- If I lose, light up red
- If you win, light up a chasing light, if I win the servo dances